

HOMEWORK 8

ELEMENTARY LINEAR ALGEBRA

(MATH 2250-03), SPRING, 2018

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SECTION 5.1 - EIGENVECTORS AND EIGENVALUES

1. Is $\lambda = 2$ an eigenvalue of $\begin{bmatrix} 3 & 2 \\ 3 & 8 \end{bmatrix}$? Why or why not?

To determine whether or not $\lambda = 2$ is an eigenvalue of $A = \begin{bmatrix} 3 & 2 \\ 3 & 8 \end{bmatrix}$, we need only determine whether or not there exists a corresponding eigenvector. That is, we must find a nonzero vector $v \in \mathbb{R}^2$ such that $Av = \lambda v$. Re-arranging, notice that $Av = \lambda v$ is equivalent to $(A - \lambda I)v = 0$. So, if we find a nonzero vector $v \in \text{Nul}(A - \lambda I)$, then we have produced an eigenvector corresponding to λ . With this in mind, observe that

$$\begin{aligned} A - \lambda I &= \begin{bmatrix} 3 & 2 \\ 3 & 8 \end{bmatrix} - 2 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 3-2 & 2 \\ 3 & 8-2 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}. \end{aligned}$$

To find $\text{Nul}(A - \lambda I)$, we must solve the homogeneous linear system $(A - \lambda I)x = 0$. Row reducing, we have

$$\begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix} \xrightarrow{\text{Row Reduce}} \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}.$$

So, the homogeneous linear system $(A - \lambda I)x = 0$ is equivalent to the system $x_1 + 2x_2 = 0$, and the parametric vector form of the solution is

$$x = \begin{bmatrix} -2x_2 \\ x_2 \end{bmatrix} = x_2 \begin{bmatrix} -2 \\ 1 \end{bmatrix}.$$

Since this then says that $\text{Nul}(A - \lambda I) = \text{span}\left\{\begin{bmatrix} -2 \\ 1 \end{bmatrix}\right\}$, the argument at the beginning of the problem tells us that any nonzero vector $v \in \text{span}\left\{\begin{bmatrix} -2 \\ 1 \end{bmatrix}\right\}$ is an eigenvector corresponding to the eigenvalue $\lambda = 2$. Since such a nonzero vector certainly exists (i.e. as $\text{Nul}(A - \lambda I)$ is not the trivial subspace $\{0\}$), we may conclude that $\lambda = 2$ is indeed an eigenvalue of $\begin{bmatrix} 3 & 2 \\ 3 & 8 \end{bmatrix}$.

2. Is $\begin{bmatrix} 4 \\ -3 \\ 1 \end{bmatrix}$ an eigenvector of $\begin{bmatrix} 3 & 7 & 9 \\ -4 & -5 & 1 \\ 2 & 4 & 4 \end{bmatrix}$? If so, find the eigenvalue.

To determine if $v = \begin{bmatrix} 4 \\ -3 \\ 1 \end{bmatrix}$ an eigenvector of $A = \begin{bmatrix} 3 & 7 & 9 \\ -4 & -5 & 1 \\ 2 & 4 & 4 \end{bmatrix}$, we begin by computing

$$Av = \begin{bmatrix} 3 & 7 & 9 \\ -4 & -5 & 1 \\ 2 & 4 & 4 \end{bmatrix} \begin{bmatrix} 4 \\ -3 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}.$$

Since $\lambda = 0$ satisfies $\lambda \begin{bmatrix} 4 \\ -3 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$, and since $v = \begin{bmatrix} 4 \\ -3 \\ 1 \end{bmatrix}$ is a nonzero vector, we have found a

nonzero vector v and a scalar λ such that $Av = \lambda v$. Therefore, we may conclude that $\begin{bmatrix} 4 \\ -3 \\ 1 \end{bmatrix}$ is an

eigenvector of $\begin{bmatrix} 3 & 7 & 9 \\ -4 & -5 & 1 \\ 2 & 4 & 4 \end{bmatrix}$ which corresponds to the eigenvalue $\lambda = 0$.

3. Let $A = \begin{bmatrix} 4 & -2 \\ -3 & 9 \end{bmatrix}$. Find a basis for the eigenspace corresponding to the eigenvalue $\lambda = 3$.

As in Problem 1, to find a basis for the eigenspace corresponding to the eigenvalue $\lambda = 3$, we need only find a basis for $\text{Nul}(A - \lambda I)$. So, we first compute

$$\begin{aligned} A - \lambda I &= \begin{bmatrix} 4 & -2 \\ -3 & 9 \end{bmatrix} - 3 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 4-3 & -2 \\ -3 & 9-3 \end{bmatrix} \\ &= \begin{bmatrix} 1 & -2 \\ -3 & 6 \end{bmatrix}. \end{aligned}$$

To find $\text{Nul}(A - \lambda I)$, we must now solve the homogeneous linear system $(A - \lambda I)x = 0$. Row reducing, we have

$$\begin{bmatrix} 1 & -2 \\ -3 & 6 \end{bmatrix} \xrightarrow{\text{Row Reduce}} \begin{bmatrix} 1 & -2 \\ 0 & 0 \end{bmatrix}.$$

So, the homogeneous linear system $(A - \lambda I)x = 0$ is equivalent to the system $x_1 - 2x_2 = 0$, and the parametric vector form of the solution is

$$x = \begin{bmatrix} 2x_2 \\ x_2 \end{bmatrix} = x_2 \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

Since this then says that $\text{Nul}(A - \lambda I) = \text{span}\left\{\begin{bmatrix} 2 \\ 1 \end{bmatrix}\right\}$, we may conclude that $\left\{\begin{bmatrix} 2 \\ 1 \end{bmatrix}\right\}$ is a basis for the eigenspace corresponding to the eigenvalue $\lambda = 3$.

4. Show that if A^2 is the zero matrix, then the only eigenvalue of A is 0.

Suppose that $A^2 = 0$ for some $n \times n$ matrix A and let λ be an arbitrary eigenvalue of A . Since we have supposed that λ is an eigenvalue of A , then by definition it must be true that $Av = \lambda v$ for some nonzero vector v . So, we have

$$\begin{aligned} 0 &= A^2v = AA v \\ &= A(\lambda v) \\ &= \lambda Av = \lambda^2 v. \end{aligned}$$

But, since v is not the zero vector, the statement $0 = \lambda^2 v$ then tells us that $\lambda = 0$ must hold. So, since λ was taken as an arbitrary eigenvalue of A , our conclusion that $\lambda = 0$ holds irrespective of our choice of eigenvalue λ . Therefore, it must be the case that $\lambda = 0$ is the only eigenvalue of A .

SECTION 5.2 - THE CHARACTERISTIC EQUATION

5. Find the characteristic polynomial and the eigenvalues of the matrix $\begin{bmatrix} 2 & 7 \\ 7 & 2 \end{bmatrix}$.

To find the characteristic polynomial of the matrix $\begin{bmatrix} 2 & 7 \\ 7 & 2 \end{bmatrix}$, recall that the characteristic polynomial of a matrix A is simply the polynomial $\det(A - \lambda I)$. So, we first compute:

$$\begin{bmatrix} 2 & 7 \\ 7 & 2 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} (2-\lambda) & 7 \\ 7 & (2-\lambda) \end{bmatrix}.$$

Now, we compute the determinate of the matrix above:

$$\begin{aligned} \det \left(\begin{bmatrix} (2-\lambda) & 7 \\ 7 & (2-\lambda) \end{bmatrix} \right) &= (2-\lambda)(2-\lambda) - (7)(7) \\ &= 4 - 4\lambda + \lambda^2 - 49 \\ &= \lambda^2 - 4\lambda - 45. \end{aligned}$$

So, the characteristic polynomial of the matrix $\begin{bmatrix} 2 & 7 \\ 7 & 2 \end{bmatrix}$ is $\lambda^2 - 4\lambda - 45$. To find the eigenvalues of this matrix, we need only solve $\lambda^2 - 4\lambda - 45 = 0$. Factoring, we have

$$\lambda^2 - 4\lambda - 45 = (\lambda - 9)(\lambda + 5).$$

So, the eigenvalues of the matrix $\begin{bmatrix} 2 & 7 \\ 7 & 2 \end{bmatrix}$ are $\lambda_1 = 9$ and $\lambda_2 = -5$.

6. Let A be an $n \times n$ matrix and suppose A has n real eigenvalues, $\lambda_1, \dots, \lambda_n$, repeated according to their multiplicities, so that

$$\det(A - \lambda I) = (\lambda_1 - \lambda)(\lambda_2 - \lambda) \dots (\lambda_n - \lambda).$$

Show that $\det(A)$ is equal to the product of the eigenvalues of A repeated according to their multiplicities, i.e. show that

$$\det(A) = \lambda_1 \lambda_2 \dots \lambda_n.$$

To show that $\det(A) = \lambda_1 \lambda_2 \dots \lambda_n$, let us observe that $\det(A - \lambda I) = \det(A)$ when $\lambda = 0$. So, setting $\lambda = 0$ and computing, we have

$$\begin{aligned} \det(A) &= \det(A - (0)I) \\ &= (\lambda_1 - 0)(\lambda_2 - 0) \dots (\lambda_n - 0) \\ &= \lambda_1 \lambda_2 \dots \lambda_n. \end{aligned}$$

So, as desired, we have shown that $\det(A) = \lambda_1 \lambda_2 \dots \lambda_n$.

Side note: While it may or may not be the case that $\lambda = 0$ is an eigenvalue of A , it is important to note that *we are not making this assumption* when we chose to evaluate the characteristic polynomial $\det(A - \lambda I)$ at the value $\lambda = 0$. That is, computing $\det(A - \lambda I)$ by plugging a value in for λ is *not the same as saying that the value you plugged in is an eigenvalue of A* . Namely, since λ is an eigenvalue of A if and only if $\det(A - \lambda I) = 0$ (that is, eigenvalues are roots of the characteristic polynomial), it is clearly true that $\lambda_1 \lambda_2 \dots \lambda_n = 0$ if and only if $\lambda_i = 0$ for some $i \in \{1, 2, \dots, n\}$. So, if $\lambda_1 \lambda_2 \dots \lambda_n \neq 0$, this simply means that 0 is not an eigenvalue of A .

SECTION 5.3 - DIAGONALIZATION

7. The matrix $A = \begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$ may be factored in the form $A = PDP^{-1}$ as

$$\begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 0 & -1 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} 5 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1/4 & 1/2 & 1/4 \\ 1/4 & 1/2 & -3/4 \\ 1/4 & -1/2 & 1/4 \end{bmatrix}.$$

Use the Diagonalization Theorem to find the eigenvalues of A and a basis for each eigenspace.

Recall that the Diagonalization Theorem states that a matrix A satisfies $A = PDP^{-1}$ for some invertible matrix P and diagonal matrix D if and only if the columns of P are linearly independent eigenvectors of A . Moreover, it also states that, in such cases, the diagonal entries of D are eigenvalues of A that correspond, respectively, to the eigenvectors in P .

So, with this in mind, since we know that

$$A = PDP^{-1}$$

$$\begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 0 & -1 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} 5 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1/4 & 1/2 & 1/4 \\ 1/4 & 1/2 & -3/4 \\ 1/4 & -1/2 & 1/4 \end{bmatrix}$$

it must be the case that the vectors

$$v_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \quad v_3 = \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix}$$

are linearly independent eigenvectors of A and the scalars

$$\lambda_1 = 5, \quad \text{and} \quad \lambda_2 = 1$$

are eigenvalues of A . Moreover, since the first entry on the diagonal of D corresponds to the eigenvalue $\lambda = 5$, it must be the case that the first column of P (i.e. v_1) is an eigenvector of A corresponding to the eigenvalue $\lambda = 5$. Hence,

$$\{v_1\} = \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}$$

is a basis for the eigenspace of A corresponding to the eigenvalue $\lambda = 5$. By similar logic, as the second and third entries on the diagonal of D correspond to the eigenvalue $\lambda = 1$ of A , the second and third columns of P (i.e. v_2 and v_3) are linearly independent eigenvectors of A corresponding to the eigenvalue $\lambda = 1$. Hence,

$$\{v_2, v_3\} = \left\{ \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix} \right\}$$

is a basis for the eigenspace of A corresponding to the eigenvalue $\lambda = 1$.

8. Suppose A has n linearly independent eigenvectors. Prove that A^T also has n linearly independent eigenvectors. (HINT: Use the Diagonalization Theorem)

Suppose that A has n linearly independent eigenvectors, and denote them by $v_1, \dots, v_n$. Then, by the Diagonalization Theorem, we have

$$A = PDP^{-1} = \begin{bmatrix} v_1 & \dots & v_n \end{bmatrix} D \begin{bmatrix} v_1 & \dots & v_n \end{bmatrix}^{-1}$$

where D is a diagonal matrix whose i^{th} entry on the diagonal is the eigenvalue of A corresponding to the eigenvector v_i . So, since $D^T = D$ for any diagonal matrix D and $(M^{-1})^T = (M^T)^{-1}$ for any invertible matrix M , we have

$$\begin{aligned} A^T &= (PDP^{-1})^T \\ &= (P^{-1})^T D^T P^T \\ &= (P^T)^{-1} D P^T. \end{aligned}$$

Writing $R = (P^T)^{-1}$, this then becomes $A^T = RDR^{-1}$. Moreover, as R is an invertible matrix (as the transpose of an invertible matrix is always invertible), the Diagonalization Theorem yields that the columns of R must be linearly independent eigenvectors of A . Therefore, since R has n columns, we may conclude that A^T has n linearly independent eigenvectors as well.